Detailed Summary of the Local-Smoothing and Exponential-Time-Hypothesis Reports

Revision 3: Contributions from the 13-Volume Ram Calculus Corpus, Proved Improvements, Computational Evidence, and Open Questions

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Abstract

This report summarizes the third revisions of the Ram Calculus studies of the OpenAI manuscripts on Euclidean wave local smoothing and the Exponential Time Hypothesis (ETH). It distinguishes the results newly derived using the 13-volume Ram MathWorld supplement from material retained from the earlier reports. For local smoothing, the principal additions are exact discarded-mode memory, a sharp quadratic retained-channel error bound, a unitary residual certificate, a metric-compatible moving-packet error estimate, and a conditional cofactor-weighted correction of a linear observable. A proved unstable closure and manufactured tests identify important limitations. For ETH, the additions are target-identifiability and precision criteria, sharp finite degree-dependent two-adic precision losses, and certified orbit and quotient reductions. Previously measured Boolean-jet speedups are retained but are not new revision-3 timings. The improvements concern explicit representations, conditional guarantees, finite arithmetic requirements, and structured computations. Neither report proves a stronger unrestricted local-smoothing estimate or a better general SAT running-time exponent. The summary explains the mathematical content, its relation to conventional methods, the evidence actually obtained, and the work still needed to affect the source papers' full algorithms or headline theorems.

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1 Scope, sources, and interpretation of contributions

1.1 The two reports summarized

The primary sources for this summary are:

1. *Ram Calculus and Local Smoothing for the Euclidean Wave Equation*, revision 3, 43 pages [1]; its new corpus-derived results are concentrated in Section 20.
2. *Ram Calculus and the Exponential Time Hypothesis*, revision 3, 45 pages [2]; its new corpus-derived results are concentrated in Section 17, while Section 16 retains the implementation and benchmark study from revision 2.

The summary is a reading guide and synthesis of those reports, not a new execution of the experiments or an independent certification of the entire OpenAI proofs. The source manuscripts are dated October 25, 2026 [4, 5]. The v3 provenance records pin the examined repository to commit

`42907c9e999cc00d6abe7a50bfc1023250edd1b9.`

The v3 checks record matching public PDFs and matching inspected source files on October 26. Statements about a source theorem below remain *statements of that source* unless the Ram report supplies its own proof of the particular assertion. This document makes no claim about subsequent changes or mathematical-community acceptance.

1.2 What was learned from the 13 volumes

The Ram MathWorld supplement contains 627 topics: 428 associated with MathWorld topics and 199 Ram-specific entries [3]. The v3 work inventoried and screened all 627 entries, then performed targeted mathematical analysis of the relevant material. It did *not* freshly prove or audit every statement in all 13 books.

Report	Explicit crosswalk	Supporting background	Not selected
Local smoothing	59	140	428
ETH	32	81	514

Table 1: Screening dispositions in the v3 topic register. Each row totals 627. These are not counts of new theorems; the two reports' selections can overlap.

The volumes made several previously implicit connections explicit: projection loss can be represented by memory; a moving representation requires a connection and a physical metric; recovery of one target is different from recovery of every coefficient; and a compressed state must be closed under every operation that follows it. The v3 reports specialize these principles to the actual questions raised by the two source papers.

Three different meanings of "new" must be distinguished:

1. **New to the revised report:** a derivation, estimate, example, or failure test absent from its predecessor.
2. **An explicit application-level deduction:** a result obtained by joining corpus statements with the particular wave or counting problem. This can be useful even when the ingredients are classical.

3. **A historically new mathematical theorem:** a priority claim requiring a substantially broader literature investigation. Neither v3 report establishes such a claim for projection memory, Galerkin methods, adjoint estimation, Hermite interpolation, module membership, or symmetry reduction as general techniques.

The strongest defensible reading of the contribution is the second one: explicit, checked, problem-specific refinements with stated assumptions and boundaries, rather than an assertion that conventional mathematics could not derive the same identities.

2 Local smoothing: the source problem and retained results

2.1 The headline estimate is not improved by revision 3

For the half-wave propagator $U(t) = e^{-\mathrm{i}t|D|}$ in spatial dimension $n \geq 2$, set $p_c = 2n/(n-1)$. The inspected source states a critical frequency-localized estimate of the form

$$\|U(t)\varphi(D/N)f\|_{L^{p_c}(\mathbb{R}^n \times [1,2])} \leq C[\log(2+N)]^{2/p_c} \|f\|_{L^{p_c}} . \tag{1}$$

Here φ localizes to a frequency annulus and the constant is uniform in the large frequency parameter N. The source's focusing example requires at least a logarithmic factor of order $(\log N)^{1/p_c}$. The gap between these powers is not closed by the Ram report. Nor does v3 prove a better unrestricted smoothing exponent.

This distinction matters because a useful finite-dimensional error bound does not automatically prove an estimate uniform over all frequencies and all admissible data. The source uses analytic and geometric arguments whose constants, spatial localization, boundary behavior, and critical norms must remain controlled in precisely that uniform limit.

2.2 The auxiliary PDE and the Ram connection

The auxiliary problem is a quaternionic Moore-determinant Dirichlet equation, schematically

$$D_w = H_{\mathrm{fib}}(B_0 + w), \qquad \det_{\mathrm{M}} D_u = \lambda\mu, \qquad D_u > 0. \tag{2}$$

The fiber Hessian, rank-one background B_0, positive branch, periodic directions, and Dirichlet wall are part of the problem, not dispensable notation. In the source argument the wall also moves with the outer parameter. This is not simply another wave equation, and it is not a constant-coefficient Ram partial differential operator with an immediate universal inverse.

The earlier Ram report already supplied exact null-coordinate and Gaussian-semigroup reformulations, boundary-aware cofactor Newton equations, conditional residual-to-solution and residual-to-energy estimates, and an off-solution energy identity. The latter isolates a product of the determinant residual and a spacetime Schur complement; thus a small determinant residual alone need not certify the entire source energy argument. Rank-one models, a planar weak inversion example, frame reconstruction budgets, phase fidelity, and logarithmic obstructions were also retained. These are valuable parts of v3, but they are not additions first obtained from the 13-volume re-examination.

2.3 Quaternionic TR60 and TR3: what they do and do not supply

The corrected quaternionic Ram Calculus in TR60 and its revised foundational treatment supplies ordered multiplication rules and a real-block interpretation. TR3 in the RMT-extensions series supplies constrained-channel observation, calibration, and physical Gram-metric tools. These were already used in the earlier local-smoothing report. Their significance is to prevent invalid commutations, metric changes, and assertions of invertibility after information has been discarded.

They do not prove that an arbitrary nonlinear quaternionic determinant equation can be diagonalized by a scalar transform. In particular, one cannot commute general quaternionic phases or ignore the boundary conditions. The new v3 cofactor comparison uses positive quaternionic Hermitian geometry explicitly. The new memory theorem, by contrast, is stated for a complex Hilbert-space self-adjoint evolution; applying it to a different algebra requires a valid realization of those hypotheses.

An identifier caution is useful: the corpus entry **R60** on physical wave Gram energy and the technical report **TR60** on quaternionic Ram Calculus are different documents.

3 Local smoothing: new deductions from the 13 volumes

3.1 The most consequential corpus connections

Corpus entries	Principle made explicit	Use in the v3 wave report
R22 (Vol. 5), R117 and R198 (Vols. 11, 13)	Projection leakage, quotient closure, intertwining defects	Distinguishes an exact compressed evolution from a discarded-mode approximation.
R199 (Vol. 13)	Dynamic Kron reduction, memory and finite lag moments	Gives an exact wave-memory equation, a sharp retained-mode bound, and a trajectory certificate.
R60 and R88 (Vol. 6), R148 and R192 (Vols. 12, 13)	Physical Gram metrics, adjoints, moving packets and compatible derivatives	Leads to a moving-frame connection and a gauge-invariant leakage bound.
R35 (Vol. 5), R69 (Vol. 6), M177–M179 (Vol. 4)	Weak residuals, goal budgets, Green identity, Poincare and Lax–Milgram structure	Leads to a signed cofactor goal correction with computable bounds when sufficient certificates are available.

R69 is not itself a theorem for the nonlinear auxiliary PDE: its goal-budget perspective motivates a separate PDE derivation. Likewise, a finite memory-moment identity in R199 is not by itself a stable trajectory approximation. The new proofs address these missing links.

3.2 Exact discarded-mode memory

Let H be bounded and self-adjoint on a complex Hilbert space, P an orthogonal projection, and $Q = I - P$. In the retained/discarded splitting, write

$$H = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}, \qquad u(t) = e^{-\mathrm{i}tH}u_0 = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}, \qquad \beta = \|B\|\,. \tag{3}$$

Solving the discarded equation exactly gives

$$y(t) = e^{-\mathrm{i}tD} y_0 - \mathrm{i} \int_0^t e^{-\mathrm{i}(t-s)D} B^* x(s)\, \mathrm{d}s. \tag{4}$$

Substitution into the retained equation yields

$$x'(t) = -\mathrm{i}Ax(t) - \mathrm{i}Be^{-\mathrm{i}tD} y_0 - \int_0^t K(t-s)x(s)\, \mathrm{d}s, \tag{5}$$

$$K(s) = Be^{-\mathrm{i}sD} B^*. \tag{6}$$

The kernel satisfies $K(0) = BB^* \geq 0$, $K(s)^* = K(-s)$, and $\|K(s)\| \leq \beta^2$. It need not decay and need not be positive for all times. The retained dynamics is memory-free for every initial state only when $B = 0$, namely when the retained space is invariant under H.

The contribution is an exact account of what a reduced wave representation has discarded. Rather than declaring the compressed coefficients to obey the same closed equation, it identifies both a history term and a forcing term from the initially unobserved state. This is a wave specialization of classical projection/elimination mathematics, organized through the corpus closure and memory framework.

3.3 A sharp quadratic improvement for the retained channel

The memory-free approximation is $x_M(t) = e^{-\mathrm{i}tA} x_0$. The proved bound is

$$\boxed{\|x(t) - x_M(t)\| \leq t\beta \|y_0\| + \frac{t^2\beta^2}{2} \|x_0\|.} \tag{7}$$

When $y_0 = 0$, leakage has to leave the retained space and return before affecting x. That two-step mechanism gives the quadratic factor $t^2\beta^2/2$, sharper at short time than a generic first-order full-state estimate. It does *not* mean that the omitted state y is only second order: it is generally first order.

The two-mode example

$$H = \nu I + \begin{pmatrix} 0 & \omega \\ \omega & 0 \end{pmatrix}, \quad \nu > \omega > 0, \quad u_0 = (1, 0)^T \tag{8}$$

has $x = e^{-\mathrm{i}\nu t} \cos(\omega t)$ and $y = -\mathrm{i}e^{-\mathrm{i}\nu t} \sin(\omega t)$. Thus the retained error is $1 - \cos(\omega t)$, asymptotic to $\omega^2 t^2/2$, while the full error against the naive lift $(x_M, 0)$ is $2\,|\sin(\omega t/2)|$. The constant $1/2$ is sharp at small time. If $y_0 \neq 0$, a first-order retained error is unavoidable even when $x_0 = 0$.

3.4 A full-memory residual certificate without an exponential factor

For a differentiable trial $\widehat{x}$ with the correct initial value, define $\widehat{y}$ by the exact lift (4), using $\widehat{x}$ in the integral. Set

$$r(t) = \widehat{x}'(t) + \mathrm{i}A\widehat{x}(t) + \mathrm{i}Be^{-\mathrm{i}tD} y_0 + (K * \widehat{x})(t). \tag{9}$$

Then

$$\boxed{\|u(t) - (\widehat{x}(t), \widehat{y}(t))\| \leq \int_0^t \|r(s)\|\, \mathrm{d}s.} \tag{10}$$

This follows from the unitary full evolution and Duhamel's formula. It avoids an unnecessary Gronwall exponential because the relevant propagator has norm one. An initial mismatch adds its full-state norm. Kernel approximation, time quadrature and calibration errors have to be added to the residual; they are not absorbed for free.

For a Taylor approximation of $\widehat{x}(t-s)$ through order N, with $\sup\left\|\widehat{x}^{(N+1)}\right\| \le F_{N+1}$, the kernel-moment remainder is bounded at time t by

$$\frac{\beta^2 F_{N+1} t^{N+2}}{(N+1)!(N+2)}. \tag{11}$$

If this is the only residual defect, its integrated contribution up to T is at most

$$\frac{\beta^2 F_{N+1} T^{N+3}}{(N+1)!(N+2)(N+3)}. \tag{12}$$

This supplies a precise bridge from a moment approximation to a trajectory error budget. It requires an actual derivative bound for the computed trial. It does not prove that a truncated moment equation is stable, nor that exact memory evaluation is inexpensive.

3.5 A proved failure: finite memory closure can be unstable

After removing the common carrier in the two-mode example, the exact retained amplitude obeys $a' = -\omega^2 \int_0^t a(s)\,\mathrm{d}s$, hence $a = \cos(\omega t)$ for the specified initial data. A second-order Taylor closure at the current time instead gives

$$\frac{\omega^2 t^3}{6} a'' + \left(1 - \frac{\omega^2 t^2}{2}\right) a' + \omega^2 t a = 0. \tag{13}$$

Its asymptotic fundamental solutions are $t^{2\pm \mathrm{i}\sqrt{2}}(1 + O(t^{-2}))$; nonzero real solutions have an unbounded, oscillatory t^2 envelope. The v3 proof uses a convergent large-time series, not merely numerical growth.

For $\omega = 2$, initialized at $t = 0.5$ from the exact cosine and its derivative, the computed closure reaches approximately 74.0903 at $t = 10$, while the exact amplitude is 0.408082. Halving the RK4 step from 0.001 to 0.0005 changes the reported sampled values by at most about 1.95×10^{-11}. This supports the calculation, while the asymptotic argument establishes the mathematical instability. The constructive lesson is to use the full residual certificate and structure-preserving approximations, not to infer stable dynamics from a formally accurate finite moment expansion.

3.6 Moving-packet synthesis: conservation and accuracy are separate

Let $V(t) : \mathbb{C}^m \to \mathcal{H}$ be an injective C^1 synthesis map with $G(t) = V(t)^* V(t) \ge g_0 I$. Define

$$P = V G^{-1} V^*, \qquad W = V' + \mathrm{i} H V, \tag{14}$$

$$\Gamma = G^{-1} V^* W, \qquad L = (I - P) W. \tag{15}$$

The reduced equation $c' = -\Gamma c$ produces $v = Vc$ satisfying

$$v' + \mathrm{i} H v = Lc, \qquad V^*(v' + \mathrm{i} H v) = 0, \qquad (c^* G c)' = 0. \tag{16}$$

The connection includes the moving-frame term V'. With $E_0 = c(0)^* G(0) c(0)$ and $\eta(t) = \left\| L(t) G(t)^{-1/2} \right\|$, the accuracy certificate is

$$\boxed{\|u(t) - v(t)\| \le \|u_0 - v_0\| + \sqrt{E_0} \int_0^t \eta(s)\, \mathrm{d}s.} \tag{17}$$

Under $V \mapsto VS$, $c \mapsto S^{-1}c$ for invertible time-dependent S, the connection transforms as $\Gamma \mapsto S^{-1}\Gamma S + S^{-1}S'$. Physical energy and η are invariant. A Euclidean norm on arbitrary coefficients is not the correct substitute for the Gram metric.

An explicit example uses $H = \nu I$ and

$$V(t) = \begin{pmatrix} e^{\gamma t}\cos t & 0 \\ e^{\gamma t}\sin t & 0 \\ 0 & 1 \end{pmatrix}. \tag{18}$$

For the indicated first-channel initial state, the compatible solution is $v = e^{-\mathrm{i}\nu t}(\cos t, \sin t, 0)^T$. Its energy is exactly one, but its error from the true stationary-direction evolution is $2\,|\sin(t/2)|$, approximately 0.958851 at $t = 1$. Omitting V' additionally introduces spurious energy growth $e^{2\gamma t}$, equal to about 1.82212 for $\gamma = 0.3, t = 1$. Thus correct energy conservation prevents one defect, while the leakage certificate quantifies the remaining approximation error.

The passage from this L^2 statement to the critical spacetime norm is conditional. If H preserves one fixed Fourier annulus and the initial data and every range of $V(t)$ stay in that annulus, Bernstein conversion introduces a factor of order $N^{1/2}T^{1/p_c}$. This does not yield (1) with an improved uniform logarithm. Spatial tapering can create frequency tails and invalidate an unqualified annulus argument.

3.7 A signed cofactor correction for a linear observable

This result concerns one fixed, smooth, compact periodic fiber with a nonempty Dirichlet wall. It does not silently replace the source's moving-boundary problem. Let u, v have sufficient C^3 regularity, the same zero Dirichlet trace, and positive fiber Hessians D_u, D_v. Assume u solves (2). On the real weak space $\mathcal{V} = H^1_{D,\mathrm{per}}$ define

$$C_w = \lambda^{-1} \operatorname{cof} D_w, \tag{19}$$

$$a_w(p, q) = \int \operatorname{Re}\left[(G_{\mathrm{fib}} p)^* C_w (G_{\mathrm{fib}} q) \right], \tag{20}$$

$$R_v(q) = \int (\lambda^{-1} \det_{\mathrm{M}} D_v - \mu) q. \tag{21}$$

Writing $e = u - v$ and $\overline{a} = \int_0^1 a_{v+te}\, \mathrm{d}t$, the mean cofactor identity has the sign

$$R_v(q) = \overline{a}(e, q), \qquad \|R_v\|_{\overline{a},*} = \|e\|_{\overline{a}}. \tag{22}$$

Assume certified comparisons $m a_v \le \overline{a} \le M a_v$, with $0 < m \le 1 \le M$. Set $r = \|R_v\|_{a_v,*}$ and $\eta_c = \max(1 - m, M - 1)$. Then

$$r/M \le \|e\|_{a_v} \le r/m. \tag{23}$$

For a bounded linear observable $\ell \in \mathcal{V}^*$ and an approximate adjoint $\widehat{z} \in \mathcal{V}$, put $d_\ell = \|\ell - a_v(\widehat{z}, \cdot)\|_{a_v,*}$. The new goal bound is

$$\boxed{|\ell(u) - \ell(v) - R_v(\widehat{z})| \le \frac{r}{m}\left(d_\ell + \eta_c \|\widehat{z}\|_{a_v} \right).} \tag{24}$$

It suggests the corrected prediction $\ell(v) + R_v(\widehat{z})$, with an explicit uncertainty interval. An exact frozen-cofactor adjoint makes $d_\ell = 0$; cofactor variation still remains unless the mean and frozen forms coincide.

The proof subtracts the adjoint identity from (22), splits the difference into adjoint defect and cofactor variation, and applies the energy bound. It adds a signed, observable-specific correction to a state-norm residual certificate. This is an adjoint/dual-weighted-residual mechanism specialized to the quaternionic cofactor problem, not a new general theory of adjoint errors.

A sufficient comparison certificate is

$$(1-\delta)D_v \le D_u \le (1+\delta)D_v, \quad 0 \le \delta < 1, \qquad m = (1-\delta)^{d-1}, \quad M = (1+\delta)^{d-1}, \tag{25}$$

where d is the size of the fiber Hessian. The quaternionic argument uses the determinant of the positive compression to the orthogonal complement of a vector, of dimension $d-1$.

What is needed to make the certificate computable. The identity containing u is exact but is not by itself a numerical estimator. One needs a continuum enclosure establishing positivity and relative Hessian bounds, and certified dual residual bounds. Positivity at sampled points, an H^1 error estimate, or a finite vector of residual tests does not supply these facts. An equilibrated flux with exact divergence equal to the residual density provides an upper bound of the form $r \le \left\| A_v^{-1/2} q \right\|_{L^2}$; an inexact divergence adds its own dual residual. General point evaluation is not bounded on H^1 in the relevant dimensions. The result is not a certificate for every nonlinear energy, every moving-wall contribution, or the spacetime Schur complement.

A diagnostic example with exact correction. In a scalar cofactor disk model, let $u = (\rho^2-1)/4$, $v = (1-\varepsilon)u$, and $\ell(w) = \int w$. The cofactor is one, $m = M = 1$, and the adjoint is $z = (1-\rho^2)/4$. Then

$$\ell(u-v) = -\varepsilon\pi/8 = R_v(z). \tag{26}$$

For $\widehat{z} = (1-\alpha)z$, the remaining error and its bound are both $\alpha\varepsilon\pi/8$. With $\varepsilon = 0.2, \alpha = 0.1$, the true goal error is approximately -0.0785398, the correction is -0.0706858, and the remainder is 0.00785398. This checks the sign and the bound's sharpness in the scalar model. It is not a validation of the entire nonlinear fiber calculation in the source.

4 Local smoothing: evidence, benefits, and open work

4.1 What the v3 tests actually establish

The recorded checks include 320 matrix/time cases, obtained from 80 positive Hermitian 7×7 matrices at four times. The maximum ratio of retained-channel error to (7)'s zero-hidden-state bound was about 0.958571. Two-mode examples check the sharp small-time constant and the first-order effect of a nonzero hidden state.

A manufactured lifted trial for the two-mode matrix with $\nu = 2, \omega = 1$ uses $\widehat{x} = e^{-2it}$ and $\widehat{y} = -ite^{-2it}$ has residual norm t. Its exact full error is

$$\sqrt{(\cos t - 1)^2 + (\sin t - t)^2} \le t^2/2. \tag{27}$$

At $t = 0.5, 1, 2$, the errors are approximately 0.124134, 0.486265, and 1.78749, respectively, below 0.125, 0.5, and 2. Moving-packet checks include a nonunitary coordinate change; the reported

metric and connection discrepancies are on the order of 10^{-16}. Ninety cofactor comparisons in real positive-definite matrices of sizes 2, 3 and 4 give maximum numerical violation about 1.34×10^{-15}. These real matrices are a subset of the quaternionic setting, not a numerical proof covering every quaternionic matrix or continuum fiber.

The analytic proofs establish conditional results; these finite tests check formulas and implementations and expose representative failures. There is no measured speedup for a full nonlinear auxiliary-PDE solver, no complete source-proof computation, and no numerical evidence for a better unrestricted logarithmic loss.

4.2 Advantages and costs

The benefit is improved transparency and controllability: discarded modes have an explicit memory, retained error has a sharper short-time order, moving-frame artifacts can be separated from physical error, and an observable can be corrected with a signed estimate. Each advantage has a cost. Memory can be expensive to evaluate; derivative and tail bounds must be certified; moving frames require Gram matrices and connection terms; and cofactor goal estimates need a continuum enclosure and an adjoint solve. Near loss of positivity, the constants can deteriorate.

These benefits are not exclusive to a Ram notation. Standard projection, unitary evolution, moving Galerkin, and adjoint methods reproduce the core mechanisms. The Ram contribution is their explicit assembly with ordered quaternionic realization, representation choice, closure checks, and a shared error-budget framework.

4.3 What remains open or proposed

1. **Stable, affordable memory approximations.** A useful method must retain a certificate such as (10) while reducing the actual cost of the history term. A finite Taylor closure is not enough.
2. **Continuum and high-frequency control.** Extend the bounded operator and annular arguments with domain, tail and norm estimates that remain uniform at the frequencies relevant to the source theorem.
3. **Certified nonlinear computation.** Produce admissible fiber solutions, Hessian enclosures, equilibrated fluxes and adjoint defects on the actual geometry, including the source's moving wall where required.
4. **A stronger headline estimate.** To reduce the logarithmic loss, one must improve the analytic step responsible for that loss while respecting the focusing lower bound. The present certificates do not do so.

5 ETH: the problem, prior contribution, and experimental baseline

5.1 No auxiliary PDE is present in the inspected ETH paper

The ETH manuscript concerns discrete satisfiability/counting constructions, polynomial deformations, finite jets, interpolation, and exact arithmetic [5]. The inspected derivation does not use an auxiliary PDE. Identifying a finite derivative or jet algebra is not evidence of a PDE. Ram finite

polynomial representations are relevant, but the wave paper's quaternionic Dirichlet mechanism is not imported into this problem.

The OpenAI manuscript's claim that ETH is false is not independently certified by the Ram v3 report. The report distinguishes a finite routine in the source's Section 3 from the more elaborate recursive/tensor construction associated with its Appendix D. Improvements to the former or to a local interpolation lemma are not automatically improvements to the full algorithm or its worst-case exponent.

5.2 The retained Boolean-jet aggregation improvement

The earlier Ram analysis aggregated a Boolean-pair loop whose call count is

$$C_q(n) = \sum_{j=0}^{\min(q,n)} 2^j \binom{n}{j} \quad \text{into} \quad B_q(n) = \sum_{j=0}^{\min(q,n)} \binom{n}{j}. \tag{28}$$

Instead of revisiting every pair separately, each retained subset U carries a small extra jet variable w that collects the appropriate weights. Schematically the aggregated quantity is

$$K_U(z, w) = \sum_{x_U=1} G_x(1_U, z)(1 + w)^{|x|-|U|} \pmod{z^J,\ w^{q-|U|+1}}. \tag{29}$$

After all products have been formed, the alternating linear functional $\sum b_k w^k \mapsto \sum(-1)^k b_k$ recovers the required signed sum. Applying it too early is invalid: this final functional is not an algebra homomorphism on the truncated ring.

Aggregation changes an exact finite amount of repeated work. It does not remove the leading general $q \log(n/q)$-type exponent in the regimes analyzed, and the extra coefficient algebra has a cost. Weighted filtrations, rank-aware summaries, and exact structural special cases were also part of the earlier report, not new v3 discoveries.

5.3 Measured speedups retained from revision 2

The matched study used 21 fixtures with at most ten variables, seven fresh-worker timing samples per method, separate memory runs, and exact diagnostic checks. All 294 timed executions, 42 memory executions and 42 diagnostic executions agreed on the exact counts, all recovered jet coefficients, and the reconstructed polynomial coefficients. An independent truth-table count checked the constant term. The deformation oracle had independent enumeration and convolution but shared some row construction helpers; its independence was therefore not complete.

All 18 cases with $q > 0$ had a shorter aggregated median, with ratios original/aggregated between 1.29 and 13.38. These are finite implementation measurements, not asymptotic speedup factors. For $q = 0$, two controls were slower and one was approximately tied, consistent with there being no call-count reduction. The raw timing ranges were sometimes broad.

Whole-pipeline traced peak-memory ratios aggregated/original ranged from 0.991 to 1.033. For $q > 0$, the incremental kernel-peak ratios ranged from about 0.98 to 1.86. Thus the measured speed benefit did not imply a memory benefit. Logical coefficient capacity per ring element can grow from J to $J(q + 1)$. Process memory also includes the interpreter and imports and must not be confused with an isolated solver workspace bound.

These were independent Python implementations of the matched Section-3 routine, not measurements of an OpenAI-released full executable. The first 16 fixtures had a stronger direct factorization

control that was faster; five node-varying fixtures removed that particular degeneracy but remained small, few-clause examples. No v3 full-algorithm timing experiment was added. The revision-2 speedups should therefore remain separate from the new v3 exact arithmetic and symmetry results.

6 ETH: new corpus-derived results

6.1 The relevant entries and their actual role

Corpus entries	Transferable principle	New use in revision 3
M225 (Vol. 7), R106 and R112 (Vol. 11)	Confluent interpolation and compatible displaced jets	Exact target recovery with explicit row and precision requirements.
R69 (Vol. 6), R152 and R165 (Vol. 12)	The requested observable need not require every coefficient; precision must serve the target	An exact two-adic module criterion, not an imported real-variance estimate.
R103 and R111 (Vol. 11), R135 and R189 (Vols. 12, 13)	Degree closure, mixed derivatives and finite Boolean algebra	Degree-aware reductions with failure tests when the degree contract is violated.
M310 (Vol. 9), R117 and R198 (Vols. 11, 13)	Permutation symmetry, quotient closure and lumpability	Certified orbit aggregation and the invariant-kernel condition for valid compressed updates.

An important distinction is that Gaussian orthogonal Hermite panels (such as corpus entry R43) are not confluent Hermite interpolation. Likewise, a real variance-pruning or rounding argument is not a two-adic divisibility certificate. The v3 report explicitly separates these meanings.

6.2 Target recovery and the limit of deleting jet data

Suppose a polynomial belongs to a specified finite coefficient space, with coefficient vector $c \in \mathbb{Q}^d$. The observed data are $h = Ac$, and the desired output is ℓc. Exact target recovery is possible if and only if

$$\ker A \subseteq \ker \ell \quad \Longleftrightarrow \quad A^T \lambda = \ell^T \text{ for some } \lambda. \tag{30}$$

Then $\ell c = \lambda^T h$. A selected subset of rows is sufficient exactly when the same condition holds for that row subset. Otherwise two admissible polynomials yield the same selected data and different target values.

This can be weaker than recovering the entire coefficient vector. However, for ordinary prefix jets at distinct nonzero nodes t, with orders $0, \dots, r_t - 1$ and total number of observations $N_S = \sum_t r_t$, the constant coefficient on the full degree-D space is recoverable *only if* $N_S > D$. If $N_S \le D$, the polynomial

$$\prod_t (T - t)^{r_t} \tag{31}$$

vanishes in every observed jet but has nonzero constant term. Conversely, $N_S > D$ makes the ordinary Hermite data injective on that polynomial space. The target-only perspective therefore does not justify arbitrary deletion of prefix derivatives. A real reduction requires a smaller certified function space, other structure, or a more economical precision schedule.

6.3 An exact target-specific two-adic precision criterion

Let $\mathcal{O} = \mathbb{Z}_{(2)}$ be the rationals with odd denominator. Suppose observation h_i has an independent error in $2^{b_i}\mathcal{O}$, and the target is needed modulo 2^p. For a fixed exact decoder λ, sufficient and necessary coordinatewise precision is

$$b_i + v_2(\lambda_i) \geq p \tag{32}$$

for every nonzero weight. Allowing the decoder to vary, a safe exact linear decoder exists precisely when

$$\boxed{\ell^T \in A^T \operatorname{diag}(2^{p-b_i})\mathcal{O}^m.} \tag{33}$$

The minimum common extra precision for an identifiable target is

$$\delta_* = min\{\delta \geq 0 : 2^\delta \ell^T \in A^T\mathcal{O}^m\}. \tag{34}$$

This is a module-membership formulation of the precision problem. It can improve on bounding denominators in one canonical full-polynomial inverse. Its proof combines exact target identification with the allowed independent error lattice. It is not a claim that ordinary real least squares or small floating-point residuals certify modular exactness.

Smith-type methods over the localized scalar ring can help after suitable denominator handling. They cannot be transferred without justification to an arbitrary nilpotent jet algebra. The report applies the reasoning coefficientwise where appropriate. No general efficient optimizer of all recursive precision choices is implemented in v3.

6.4 A sharp six-channel precision result

Consider nodes -1 and 1, with scaled jets

$$h_{t,j} = 4^j \frac{F^{(j)}(t)}{j!}, \qquad j = 0, 1, 2. \tag{35}$$

Order the six observations as $(-1,0), (-1,1), (-1,2), (1,0), (1,1), (1,2)$. The goal is $F(0)$, not every coefficient of F. On the space of polynomials of degree at most d, the exact minimum common extra precision losses proved in v3 are

d	0	1	2	3	4	5
δ_*	0	1	3	4	6	7.

(36)

For odd degrees the useful decoder weights are:

$$\begin{aligned} d = 1: &\quad (1/2, 0, 0;\ 1/2, 0, 0), \\ d = 3: &\quad (1/2, 1/16, 0;\ 1/2, -1/16, 0), \\ d = 5: &\quad (1/2, 5/64, 1/128;\ 1/2, -5/64, 1/128). \end{aligned} \tag{37}$$

For example, the degree-three formula is

$$F(0) = \frac{F(-1) + F(1)}{2} + \frac{F'(-1) - F'(1)}{4}. \tag{38}$$

The coefficient $1/16$ in the scaled-jet formula is consistent with $h_{t,1} = 4F'(t)$. For the polynomial $7 - 2T + 3T^2 + T^3$, the six exact jets are $(11, -20, 0; 9, 28, 96)$, and (38) recovers 7. For even degrees, asymmetric weights can be better than symmetric ones: degree two admits $(1/4, 0, 0; 3/4, -1/8, 0)$, and degree four admits $(11/16, 1/8, 1/64; 5/16, -1/32, 0)$.

Why these losses are optimal in the declared model. For odd degrees $2m+1$, $m = 0, 1, 2$, the witness $F_m = (T+1)^m(T-1)^{m+1}$ has odd constant term while all six jets have valuation at least $1 + 3m$. The minimum valuations are 1, 4, and 7. The corresponding even-degree witnesses $1, T^2-1, (T^2-1)^2$ give 0, 3, and 6. Thus no exact decoder with a smaller common loss can work on the whole stated polynomial class. More strongly, zero and $2^{p-1}F_m$ give identical observed residues at one bit below the claimed threshold but different targets modulo 2^p. This indistinguishability argument also rules out a nonlinear decoder under the same observation and residue contract. It is not a lower bound for general SAT algorithms.

The concrete improvement over the preceding report. In the source's local lemma fixture $a = 1, L = 2, u = 1, M = 1$, the conservative extra allowance is 64. The earlier Ram canonical six-jet analysis gave 7. The source already certifies degree $M(2L-1) = 3$ in that fixture; exploiting that fact gives the sharp allowance 4 in v3:

$$64 \text{ (source)} \quad \longrightarrow \quad 7 \text{ (v2)} \quad \longrightarrow \quad 4 \text{ (v3)}. \tag{39}$$

The last reduction does not assume a new degree hypothesis for that local fixture. Obtaining the degree-one loss would require a stronger degree restriction. These numbers count extra binary precision in a specified recovery problem; they are not running-time speedup factors.

What prevents a full-algorithm conclusion. The source may construct an entire polynomial at a node from $M(2L-1)+1$ evaluations. Discarding two derivative outputs therefore need not remove evaluations that were already performed. Reduced required precision might lower arithmetic cost, but its propagation through the actual recursive lifts and child computations has not been implemented or measured. The sharper finite lemma is proved; a faster full recursive algorithm is still a proposal.

A necessary degree-contract failure test. Using the degree-one formula on T^2 returns 1 instead of 0; using the degree-three formula on T^4 returns -1; using the degree-five formula on T^6 returns 1. All three failures occur with exact arithmetic. The benefit comes from valid structural information, not from silently ignoring higher coefficients.

6.5 Orbit aggregation: a further exact reduction under full symmetry

Suppose a group $G \le S_n$ acts on both assignments and subset variables, and the entire polynomial family is equivariant:

$$p_{gx}(gX) = p_x(X). \tag{40}$$

For the already aggregated subset contribution

$$S_U = \sum_{x_U=1} w_q(|x|, |U|)p_x(1_U), \qquad w_q(h,u) = \sum_{k=0}^{\min(q-u,h-u)} (-1)^k \binom{h-u}{k}, \tag{41}$$

change of variables gives $S_{gU} = S_U$. Consequently the sum over subsets can be replaced by a sum over orbit representatives, weighted by integer orbit sizes. No division by $|G|$ is needed. For the full symmetric group, there is one orbit at each retained subset size, so only $\min(q,n)+1$ representative contributions are required.

This reduction requires symmetry of the *full computation*: factor assignments, deformation nodes, measurements, correction jets and output functional. Symmetry of the original SAT formula alone is insufficient. For a general labelled instance the applicable group can be trivial, and finding or certifying symmetries has a cost.

(n, q)	Pair calls C_q	Aggregated B_q	Orbit calls	Weighted target
$(6, 2)$	73	22	3	5,248
$(8, 3)$	577	93	4	239,616
$(10, 4)$	4,521	386	5	14,029,824

Table 4: Exact synthetic tests for $p_x(X) = (1 + |x|)(1 + \sum_i X_i)^q$. Targets are weighted sums, not SAT counts. The table reports call counts, not measured runtime gains.

The representative contributions in these tests still use direct inner enumeration. The dramatic reduction in the outer call count does not show that the full inner computation is efficient.

An asymmetric counterexample is $n = 4, q = 1$ with $p_x(X) = 1 + 3X_1 + X_2$. Its correct target is 48. Treating all singleton subsets as equivalent and using $\{1\}$ as the representative returns 112. The failure is a structural one, not a floating-point discrepancy.

6.6 Exact quotient closure: the test every compression must pass

Let J be a surjective summary map and T_a one of the fine-state updates. A coarse update $\widehat{T}_a$ with

$$JT_a = \widehat{T}_a J \tag{42}$$

exists exactly when $\ker J$ is invariant under T_a. Every update in the computation must satisfy this condition, and the requested output must factor through J. This is the precise sense in which a compressed state is sufficient.

For example, merging $(1, 0)$ and $(0, 1)$ because both have Hamming weight one is invalid if a subsequent operation multiplies by x_1. The merged states no longer have the same future behavior. A count-only agreement on one fixture also does not establish preservation of all mixed jet coefficients required later. The v3 criterion is a guardrail for proposing further compression, not a universal construction of a small exact summary.

7 ETH: benefits, limitations, and outstanding work

7.1 What is genuinely improved

Revision 3 proves a sharper local precision requirement, including matching lower bounds for the six-channel finite model. It also gives exact conditions under which target-only recovery or symmetry saves work, and conditions ruling out unjustified compressed updates. These are stronger and more operational statements than a generic claim that jets are useful.

The benefits are precision budgeting, elimination of redundant target data when justified, and exact orbit savings for certified symmetric families. The costs include rational or localized-integer linear algebra, coefficient growth, degree certification, symmetry verification, and the memory used by mixed jet coefficients. None of these costs can be omitted from a general complexity claim.

7.2 Which benefits are specific to Ram Calculus?

The Ram organization emphasizes finite jets, target observables, representation metrics, algebraic closure and explicit error or precision budgets. That organization helped expose the specific degree-aware and equivariance opportunities in this study. The underlying algebraic facts are nevertheless compatible with ordinary interpolation, linear algebra, two-adic modules, generating functions and group actions. Neither report demonstrates a separation showing that a conventional implementation of the same valid identities is less accurate or inherently slower.

7.3 The remaining tasks before a full-algorithm improvement

1. Integrate aggregation into the actual admissible construction and the recursive tensor route, rather than only the matched finite routine.
2. Propagate certified degree bounds and target precision backwards through every node, lift, and child call before doing the arithmetic.
3. Account for bit complexity and the cost of finding good exact decoders; fewer required bits do not by themselves prove a better total running time.
4. Prove that any proposed group symmetry survives the complete deformation and correction machinery, then measure representative work as well as outer call counts.
5. Test larger, nondegenerate instances with appropriate direct and conventional controls, and compare exact counts, all required coefficients, elapsed time and workspace memory.
6. If a better unrestricted SAT exponent is claimed, give a uniform worst-case analysis including every construction and recovery cost. No such improvement is proved in v3.

8 Combined assessment of progress

Result	Established improvement	Boundary that remains
Wave retained-channel estimate	Sharp quadratic short-time error when the hidden initial state is zero.	Full-state leakage remains first order; bounded self-adjoint setting and its stated hypotheses matter.
Wave full-memory certificate	Integrated residual controls the full lifted state without an artificial exponential amplification.	Exact or certified memory and an accurate residual are still needed.
Moving packets	Correct physical energy, coordinate-invariant leakage, and an explicit trajectory bound.	Conserved energy does not imply small error or a sharper critical L^p theorem.
Cofactor goal correction	Signed correction and certified remainder for a bounded linear observable on one admissible fiber.	Continuum Hessian, positivity, flux and adjoint certificates; moving-wall extensions.

Result	Established improvement	Boundary that remains
ETH target precision	Exact identifiability and precision tests; sharp local improvement from seven to four extra bits in the degree-three fixture.	Full recursive bit-cost benefit has not been demonstrated.
ETH orbit reduction	Exact reduction to orbit representatives under whole-family equivariance.	No generic symmetry or generic $q+1$-call algorithm; representative inner work can remain expensive.
ETH Boolean-jet runtime	Retained v2 measurements show finite speedups with exact count and coefficient agreement.	Small test suite; some overhead-only cases; no full-source runtime benchmark.
Source headline theorems	Better local representations, diagnostics and conditional certificates.	No stronger unrestricted wave bound, improved general SAT exponent, or independent certification of either full proof.

8.1 What the counterexamples contribute

The negative results are an important part of the progress. They separate promising methods from invalid extrapolations: a high-order moment closure can destabilize a bounded wave; a perfectly energy-conserving reduced solution can be inaccurate; target recovery cannot discard too many prefix jets on the full polynomial class; an invalid degree assumption fails even in exact arithmetic; and Hamming-weight symmetry does not survive asymmetric updates. These results prevent an apparent simplification from being mistaken for a theorem or a speedup.

8.2 Most useful next investigations

For local smoothing, the most concrete next step is a conforming solver for a fixed admissible auxiliary fiber that reports a certified state and goal error, then compares memory and moving-packet variants at matched accuracy. A separate analytical effort would be needed to control high-frequency and moving-boundary effects uniformly enough to affect the source estimate.

For ETH, the most concrete next step is to insert the degree-aware decoder and certified precision schedule into the actual recursive implementation, logging bit lengths and child-call costs. Orbit reduction should be tested only after full-family equivariance has been proved. Both tasks must retain exact arithmetic checks and compare against the strongest available conventional controls.

9 Reproducibility and conclusion

9.1 Where to find the evidence

The v3 reports and this summary reside in `C:/Users/mural/OneDrive/Documents/res-papers/OpenAI-papers`. The subdirectory `mathworld-v3-support` contains the topic relevance register, source provenance, executable tests and recorded results. Useful records include:

- `topic-relevance-register.csv` and `screening-summary.json`: the 627-entry dispositions and source crosswalks.

- `wave-refinement-results.json` and `wave-certificate-results.json`: the wave-compression, moving-frame, cofactor and failure checks.
- `eth-refinement-results.json`, `precision-lower-bound-results.json`, and `degree-table-results.json`: exact recovery, precision, orbit and negative tests.
- `public-source-provenance.json`, `input-sha256.json`, and `delivery-verification.json`: source versions and preservation/build checks.

The older measured runtime and memory evidence is documented in the ETH report's revision-2 benchmark section and its `eth-benchmark-v2` support directory. It should not be conflated with the v3 synthetic call-count tests. This summary reports those existing measurements rather than introducing new benchmark runs.

9.2 Conclusion

The 13-volume investigation contributed most by turning broad representation ideas into explicit admissibility conditions, exact identities, sharp finite bounds, and falsifiable computational contracts. For local smoothing, the most important advances are a correct memory model, a sharper retained-channel error, compatible moving frames and a signed cofactor goal certificate. For ETH, they are sharp target precision under a certified degree bound and exact conditions for orbit and quotient compression. Each result improves a clearly specified part of the investigation, and each has a boundary that the report makes explicit.

The appropriate conclusion is therefore neither "no contribution" nor "the full source problems have been improved." There are useful proved local refinements and well-defined computational opportunities. Obtaining better unrestricted analytic estimates or a faster full exact counting algorithm requires additional work not supplied by the current reports.

References

[1] M. Subbarao, *Ram Calculus and Local Smoothing for the Euclidean Wave Equation: Quaternionic Structure, Auxiliary Dirichlet Problems, and Residual-Certified Reformulations*, revision 3, October 26, 2026. `Ram-Calculus-Local-Smoothing-Euclidean-Wave-v3.tex` and PDF, especially Section 20 and the revision-3 reproducibility appendix.

[2] M. Subbarao, *Ram Calculus and the Exponential Time Hypothesis*, revision 3, October 26, 2026. `Ram-Calculus-Exponential-Time-Hypothesis-v3.tex` and PDF, especially Sections 16–17 and the revision-3 reproducibility appendix.

[3] M. Subbarao, *Ram Calculus MathWorld Supplement*, expanded Volumes 1–13, files `Ram-Calculus-MathWorld-Volume-01.tex` through `Ram-Calculus-MathWorld-Volume-13.tex`, in `C:/Users/mural/OneDrive/Documents/mathworld-supplement/RamCal-MathWorld-Volumes`. The v3 support register records individual topic identifiers and source locations. The entry labels cited in this summary are internal corpus identifiers, not claims of independent publication or mathematical priority.

[4] OpenAI mathematics repository, *Local smoothing for the Euclidean wave equation*, October 25, 2026; inspected version recorded in Section 1. https://github.com/openai/math/tree/main/preprints/Local-smoothing-for-the-Euclidean-wave-equation-October-25-2026.

[5] OpenAI mathematics repository, *The Exponential Time Hypothesis is false*, October 25, 2026; inspected version recorded in Section 1. https://github.com/openai/math/tree/main/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026.